

Ultrametric Quantum Kernels: Exact p -adic Feature Maps for Hierarchical Data

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Abstract

A quantum kernel compares data through the fidelity of an encoding, so the geometry the model can express is fixed before any training happens. Hierarchical data (taxonomies, ontologies, phylogenies) is not Euclidean but ultrametric, and we ask which quantum feature maps induce an exactly ultrametric kernel. We give two obstructions and a matching construction. First, no product feature map does, whatever its local dimensions: multiplicativity across digits limits any such encoding to a bounded resolution depth, which covers angle and computational-basis encodings. Second, and independently, the Gram matrix of any exact realisation forces a dimension of at least $L^2 / \sum_{x,y} K(x,y)$ on L points; on the p -adic tree with a geometrically decaying profile this exceeds 2^n for every $p \geq 3$, so no n -qubit encoding suffices, including the entangling IQP and ZZ maps the first argument cannot reach. Third, *path states*, superpositions over a leaf's root-to-leaf ancestor chain with amplitudes read off the target profile, realise precisely the strictly monotone ultrametric kernels, on arbitrary rooted trees, using a number of qubits within an additive constant of that lower bound. We further show that global depolarising noise cannot break ultrametricity at any rate, since it acts on distances as an increasing affine map, and we measure the shot budget that finite sampling demands. Experiments on WordNet, Gene Ontology and the NCBI taxonomy confirm exactness to machine precision and separate ultrametricity from usable resolution: a basis encoding is ultrametric while resolving nothing. We claim no quantum speedup in kernel evaluation, and say so throughout; the contribution is a constraint on encoding design and an optimal-width encoding for models where no closed form survives.

Keywords

quantum machine learning, quantum kernels, ultrametric spaces, p -adic numbers, hierarchical representation learning

1. Introduction

A quantum kernel method embeds classical data through a feature map Φ and compares two points by the fidelity of their images,

$$K(x, y) = |\langle \Phi(x) | \Phi(y) \rangle|^2. \quad (1)$$

Everything a downstream model can express about the similarity of x and y passes through this one number, and the number is fixed by the encoding alone. Whatever geometry Φ imposes on the data is imposed before any parameter is trained [1, 2]. The choice of encoding is therefore not a preprocessing detail. It is the model's inductive bias.

Much of the data to which quantum machine learning is applied is hierarchical. Taxonomies, ontologies, phylogenies, file systems and type lattices are trees, and the natural notion of distance on a tree is not Euclidean but *ultrametric*: distances obey the strong triangle inequality

$$d(x, z) \leq \max(d(x, y), d(y, z)), \quad (2)$$

which is strictly stronger than the ordinary triangle inequality. Equivalently, the points sit at the leaves of a tree and distance decreases with the depth of the lowest common ancestor. The

canonical ultrametric space is the ring of p -adic integers $\mathbb{Z}_p$, whose valuation $v_p(x - y)$ counts matching least-significant base- p digits and so measures exactly that depth.

This paper asks one question.

Which quantum feature maps induce an exactly ultrametric kernel?

We answer with two obstructions and a matching construction.

Theorem A: product encodings cannot do it. A feature map that factorises across digits induces a multiplicative kernel, and multiplicativity bounds how deep such a map can resolve the tree, whatever the local dimension: a product map is left with three kernel values, too few for a strictly monotone profile once $n \geq 2$. The argument generalises from single digits to disjoint blocks, where the resolution bound is set by the size of one block, so it constrains a class of encodings rather than a list of them. Angle encoding and computational-basis encoding are corollaries.

Theorem B: narrow encodings cannot do it either, whatever their structure. The Gram matrix of any exact realisation is positive semi-definite with unit diagonal, which forces its rank above $L^2 / \sum_{x,y} K(x,y)$ on L points. On the regular p -ary tree with a geometrically decaying profile of parameter $s > 1$, this exceeds 2^n for every $p \geq 3$ once n passes an explicit threshold: beyond it, *no* n -qubit encoding realises that kernel on $\mathbb{Z}/p^n$, entangling or not. This is what rules out IQP and ZZ feature maps, which Theorem A does not reach.

Theorem C: path states do it exactly, at essentially the optimal width. Writing $\text{anc}_i(x)$ for the depth- i ancestor of a leaf x , the state $\Phi_f(x) = \sum_i a_i |\text{anc}_i(x)\rangle$ realises *every* strictly monotone ultrametric kernel exactly, with amplitudes determined by the target profile. The construction needs $\lceil \log_2 |V| \rceil$ qubits for a tree on $|V|$ nodes, $n + 1$ non-zero amplitudes, and $O(n \log |V|)$ gates, and it matches Theorem B to within an additive constant.

Along the way we record a fourth fact with a two-line proof. Global depolarising noise acts on the induced distance as a non-decreasing affine map, so it *cannot* break the strong triangle inequality or level constancy at any error rate (Proposition 7.1). What it destroys is contrast rather than geometry. Finite sampling behaves in the opposite way and breaks the strong triangle inequality immediately, for a reason specific to ultrametric spaces that we make precise in Section 8.

What this paper does not claim

The kernel of Theorem 6.1 is $K(x,y) = f(\lambda(x,y))$, and the lowest common ancestor of two leaves is computable classically in time linear in the depth of the tree. **There is no quantum speedup in kernel evaluation here, and we claim none.** We state this in the introduction rather than leaving a reader to discover it, because it determines how the rest should be read. The contribution is threefold: a no-go that constrains current encoding practice; a dimension bound that applies to arbitrary encodings and so survives future encoding designs; and an explicit, optimal-width encoding usable as the input stage of deeper quantum models, where no closed form survives the subsequent layers.

Contributions

- C1** A no-go theorem for product feature maps (Theorem 4.1), which bounds the resolution depth of any block-product map by the size of one block and collapses to three kernel values for genuine product maps, with corollaries for angle and computational-basis encodings.

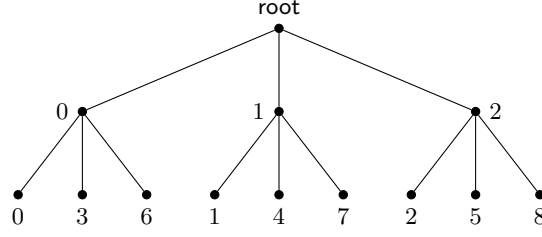


Figure 1: The regular 3-ary tree of depth 2 for Example 2.1. Internal nodes are labelled by the residue $x \bmod 3$ that defines them; leaves by $x \in \mathbb{Z}/9$. Two leaves meet at depth $v(x, y) = v_p(x - y)$: the leaves 2 and 5 share a parent, the leaves 2 and 4 share only the root.

- C2** A dimension lower bound for *arbitrary* feature maps (Theorem 5.1), which rules out every n -qubit encoding of the geometric profile on $\mathbb{Z}/p^n$ for $p \geq 3$ and n past an explicit threshold, and so covers the entangling maps Theorem A cannot reach.
- C3** An exact construction (Theorem 6.1) realising precisely the strictly monotone ultrametric kernels, with a resource count matching C2 to within an additive constant, and a converse characterising every realisation with non-negative overlaps up to isometry and phase.
- C4** A generalisation beyond the homogeneous case: the construction needs only that ancestors nest, so it applies to arbitrary rooted trees, with $\mathbb{Z}/p^n$ as the specialisation in which the profile ties to $|\cdot|_p$.
- C5** An empirical study on WordNet, Gene Ontology and the NCBI taxonomy separating ultrametricity from usable resolution, mapping the expressivity/hierarchy trade-off of a re-uploading extension, and quantifying the shot budget the construction needs.

Reproducibility. Every theorem in this paper has a corresponding numerical check, cited by test name at the point of use, and every experimental number in the text is generated from the experiment outputs rather than transcribed.

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2. Preliminaries

2.1. p -adic integers and the regular tree

Fix a prime p and an integer $n \geq 1$. Write $X = \mathbb{Z}/p^n$ and expand each $x \in X$ in base p as $x = \sum_{i=1}^n x_i p^{i-1}$ with digits $x_i \in \{0, \dots, p-1\}$. The p -adic valuation of a non-zero integer is the exponent of the largest power of p dividing it, and the p -adic absolute value is $|m|_p = p^{-v_p(m)}$. On X we use the truncated valuation

$$v(x, y) = \min(v_p(x - y), n) = \#\{\text{matching least-significant base-}p \text{ digits of } x \text{ and } y\}. \quad (3)$$

Identify X with the leaves of the regular p -ary tree of depth n , a leaf x sitting under the chain of residue classes $x \bmod p, x \bmod p^2, \dots$. Then the depth- i ancestor of x is the class $x \bmod p^i$, and $v(x, y)$ is exactly the depth of the lowest common ancestor of x and y .

Example 2.1. Take $p = 3, n = 2$, so $X = \{0, \dots, 8\}$. The leaves $x = 2$ and $y = 5$ have $x - y = -3$, hence $v_p(x - y) = 1$: they agree in the least-significant digit ($2 \equiv 5 \bmod 3$) and differ in the next, so they meet at depth 1. The leaves 2 and 4 differ already in the least-significant digit and meet only at the root, at depth 0. Figure 1 draws the tree.

2.2. Ultrametric spaces

A metric d on a set X is an *ultrametric* if it satisfies Eq. (2) for all x, y, z . Two consequences matter here. First, every triangle is isocles with its two longest sides equal, so a constant fraction of triples meets Eq. (2) *with equality*. That fact turns out to govern how ultrametricity behaves under sampling noise (Section 8). Second, the closed balls of an ultrametric space are nested or disjoint, never partially overlapping, which is precisely the combinatorics of a rooted tree.

Real hierarchical data is rarely exactly ultrametric, and quantifying the discrepancy is an active question [3]. Our concern is the complementary one: given that the target geometry *is* ultrametric, which encodings can represent it without distortion.

2.3. Quantum feature maps and fidelity kernels

A quantum feature map is a function $\Phi : X \rightarrow \mathbb{C}^D$ with $\|\Phi(x)\| = 1$, realised in practice by a state-preparation circuit on $\lceil \log_2 D \rceil$ qubits. Its fidelity kernel is Eq. (1). Writing $G_{xy} = \langle \Phi(x) | \Phi(y) \rangle$ for the Gram matrix of the feature vectors, we have $K = |G|^{\circ 2}$ entrywise, and G is positive semi-definite with $G_{xx} = 1$. Since the Schur product of positive semi-definite matrices is positive semi-definite, K is a valid kernel for any Φ .

Two families of encoding recur below. An *angle encoding* assigns one qubit per digit, $|\psi_i(a)\rangle = \cos(\theta_a/2)|0\rangle + \sin(\theta_a/2)|1\rangle$, and takes the tensor product. A *ZZ (or IQP-style) feature map* applies a data-dependent diagonal phase between layers of Hadamards, $\Phi(x) = (U_{\phi(x)} H^{\otimes n})^r |0\rangle^{\otimes n}$, with the phase carrying both single-qubit and pairwise terms [2]. The pairwise terms make the ZZ map entangling, which is exactly why Theorem 4.1 does not apply to it and Theorem 5.1 must.

3. Ultrametric Quantum Kernels

Throughout, T is a finite rooted tree whose leaves all sit at depth n (a leaf that is shallower is extended by a chain of unary nodes, which changes no lowest common ancestor). For a leaf x and $0 \leq i \leq n$ we write $\text{anc}_i(x)$ for its depth- i ancestor, so $\text{anc}_0(x)$ is the root and $\text{anc}_n(x) = x$, and $\lambda(x, y) = \max\{i : \text{anc}_i(x) = \text{anc}_i(y)\}$ for the depth at which two leaves meet. On the regular p -ary tree, $\lambda = v$ of Eq. (3).

Definition 3.1 (Ultrametric kernel). A kernel $K : X \times X \rightarrow [0, 1]$ is *ultrametric with profile* f if there is a function $f : \{0, \dots, n\} \rightarrow [0, 1]$ with $f(n) = 1$ such that $K(x, y) = f(\lambda(x, y))$ for all leaves x, y . The profile is *strictly monotone* if f is strictly increasing and $f(0) > 0$.

Being ultrametric with *some* profile is the same as being constant on each level of the lowest-common-ancestor relation, a property we call *level constancy*. Level constancy alone is not enough to make the induced distance an ultrametric: $f = (0.9, 0.1, 1)$ on a depth-2 tree is level-constant, yet two leaves meeting at depth 1 are further apart than either is from a leaf meeting them at the root. Monotonicity is what supplies the strong triangle inequality.

Lemma 3.2. *If K is ultrametric with a non-decreasing profile f satisfying $f(v) < 1$ for every $v < n$, then $d = 1 - K$ is an ultrametric on the leaves of T . Without the second hypothesis d is a pseudo-ultrametric: $f \equiv 1$ gives $d \equiv 0$.*

Proof. Lowest-common-ancestor depth satisfies $\lambda(x, z) \geq \min(\lambda(x, y), \lambda(y, z))$, because the ancestors of a leaf form a chain: if x and y agree down to depth i and y and z agree down to depth i , then x and z agree down to depth i . Applying the non-decreasing f preserves this and reverses it under $d = 1 - f$, giving Eq. (2). Non-degeneracy follows from $f(n) = 1$ and $f(v) < 1$ for $v < n$. $\square$

3.1. Two diagnostics, because one is not enough

It is tempting to certify an encoding by counting violations of Eq. (2). That test is necessary but far from sufficient, and the gap is not a technicality. Computational-basis encoding sends distinct leaves to orthogonal states, so its kernel is the identity and its induced distance is the discrete metric — which is an ultrametric, with zero violations, while collapsing the hierarchy to a single bit of information. It is the degenerate corner of Definition 3.1, with $f = \mathbf{1}[v = n]$.

We therefore report two quantities throughout.

Definition 3.3 (Level constancy and resolution depth). Let $V(K) = \{v : \lambda(x, y) = v \text{ for some pair of leaves}\}$ be the set of *populated levels*. The *level constancy* is

$$\max_{v \in V(K)} \left(\max_{\lambda(x, y) = v} K(x, y) - \min_{\lambda(x, y) = v} K(x, y) \right),$$

which vanishes exactly when K is ultrametric with some profile. Writing μ_v for the mean of K over level v , the *resolution depth* is the number of runs in the sequence $(\mu_v)_{v \in V(K)}$ after merging consecutive entries differing by at most τ ; we take $\tau = 10^{-6}$ throughout, a scale no realistic shot budget could resolve (Section 8.6). It equals $|V(K)|$ for a strictly monotone profile and 2 for the discrete metric.

Restricting to populated levels matters on padded and irregular trees, where some v may carry no pair of leaves at all. At such a v the profile is not determined by K , so Theorem 6.1 should be read as a bijection onto profiles restricted to $V(K)$.

An encoding is useful for hierarchical data only if it scores well on both: level constancy zero, resolution depth full. Theorem 6.1 exhibits the only family that does.

3.2. Product and block-product feature maps

Definition 3.4 (Product feature map). A feature map on $\mathbb{Z}/p^n$ is a *product map* if $\Phi(x) = \bigotimes_{i=1}^n |\psi_i(x_i)\rangle$ for unit vectors $|\psi_i(a)\rangle \in \mathbb{C}^{d_i}$, where factor i depends only on digit i and the local dimensions d_i are arbitrary. Writing $g_i(a, b) = |\langle \psi_i(a) | \psi_i(b) \rangle|^2$, we have $g_i(a, a) = 1$ and

$$K(x, y) = \prod_{i=1}^n g_i(x_i, y_i). \quad (4)$$

Definition 3.5 (Block-product feature map). Let $\mathcal{B}$ be a partition of the levels $\{1, \dots, n\}$ into consecutive blocks. A *block-product map* is $\Phi(x) = \bigotimes_{B \in \mathcal{B}} |\psi_B(x_B)\rangle$, where x_B is the tuple of digits in block B . Its kernel factorises across blocks in the same way. A product map is the case in which every block is a single level.

Block-product maps let a bounded number of levels interact arbitrarily while keeping the tensor structure across blocks. They are the natural strengthening of Definition 3.4, and Theorem 4.1 handles both at once. They do not, however, cover maps such as the ZZ feature map, whose all-pairs couplings leave no tensor factorisation at all; that case is Section 5.

4. A No-Go Theorem for Product Encodings

Theorem 4.1 (Product no-go). *Let Φ be a block-product feature map on $\mathbb{Z}/p^n$ (Definition 3.5) whose fidelity kernel is ultrametric with profile f . Put $v^* = \min\{v : f(v) > 0\}$ and assume $v^* < n$, and let B^* be the block containing level $v^* + 1$. Then for every block B all of whose levels exceed $\max B^*$, and all digit tuples a, b ,*

$$|\langle \psi_B(a) | \psi_B(b) \rangle| = 1. \quad (5)$$

Consequently $f(v) = 1$ for every $v \geq \max B^*$: the map resolves the tree only to depth $\max B^*$, and its resolution depth is at most $|B^*| + 2$.

For a product map every block is a single level, so $\max B^* = v^* + 1$ and f takes at most the three values $\{0, f(v^*), 1\}$. In particular no product map realises a strictly monotone profile once $n \geq 2$.

Proof. The argument uses exactly one property of the encoding: fidelity is multiplicative across tensor factors, Eq. (4).

Choose leaves x, y that agree on levels $1, \dots, v^*$, differ at level $v^* + 1$, and agree at every level above that; such a pair exists because $v^* < n$. Then $\lambda(x, y) = v^*$, and every factor except B^* contributes $g(x, x) = 1$, so

$$f(v^*) = K(x, y) = |\langle \psi_{B^*}(x_{B^*}) | \psi_{B^*}(y_{B^*}) \rangle|^2 > 0. \quad (6)$$

Now let B be a block whose levels all exceed $\max B^*$, fix digit tuples $a \neq b$ for it, and let y' agree with y everywhere except on B , where $y'_B = b$ while $x_B = a$. Because x and y' still first differ at level $v^* + 1$, and B lies strictly above B^* , we still have $\lambda(x, y') = v^*$ and therefore $K(x, y') = f(v^*)$. Multiplicativity gives

$$K(x, y') = |\langle \psi_{B^*}(x_{B^*}) | \psi_{B^*}(y_{B^*}) \rangle|^2 \cdot |\langle \psi_B(a) | \psi_B(b) \rangle|^2 = f(v^*) \cdot |\langle \psi_B(a) | \psi_B(b) \rangle|^2.$$

Since $f(v^*) > 0$ by Eq. (6), this forces $|\langle \psi_B(a) | \psi_B(b) \rangle|^2 = 1$, which is Eq. (5). Equality in Cauchy–Schwarz for unit vectors then gives $|\psi_B(b)\rangle = e^{i\theta} |\psi_B(a)\rangle$: block B carries no information about the data.

Every block beyond B^* being trivial, K depends only on the digits in blocks up to and including B^* , that is, on levels $1, \dots, \max B^*$. Two leaves agreeing down to depth $\max B^*$ therefore agree in every non-trivial factor and have $K(x, y) = 1$, so $f(v) = 1$ for all $v \geq \max B^*$. Together with $f(v) = 0$ for $v < v^*$, only the levels $v^* \leq v < \max B^*$ remain unconstrained, and they number at most $|B^*|$; the resolution depth is therefore at most $|B^*| + 2$.

For a product map $|B^*| = 1$ and $\max B^* = v^* + 1$, so the profile is 0 below v^* , equal to $f(v^*)$ at v^* , and 1 from $v^* + 1$ on: three values. A strictly monotone profile has $f(0) > 0$, hence $v^* = 0$, and needs $n + 1$ distinct values, which exceeds three whenever $n \geq 2$. $\square$

The proof turns on multiplicativity alone. It is therefore indifferent to the local dimensions d_i , to whether the local states are real or complex, and to how they were chosen, which is what makes it a statement about a class of encodings rather than about particular instances.

Remark 4.2 (The block bound is tight, so the three-value claim is genuinely about product maps). Nothing in Definition 3.5 forbids the single-block partition $\mathcal{B} = \{\{1, \dots, n\}\}$, and with one block the path state of Theorem 6.1 is itself a block-product map realising a strictly monotone profile with $n + 1$ distinct values. The theorem is consistent with this: $\max B^* = n$, so the conclusion $f(v) = 1$ for $v \geq n$ is vacuous. What the theorem constrains is how far a block-product map can resolve *beyond* the block containing the first informative level, and for genuinely factorised encodings — where blocks are small — that is a severe constraint. It is only for $|B^*| = 1$ that it collapses to three values.

Corollary 4.3. *On $\mathbb{Z}/p^n$ with $n \geq 2$, none of the following induces a strictly monotone ultrametric kernel: angle encoding with one qubit per digit; computational-basis encoding; and any first-order Pauli-Z feature map, whose phase is a sum of single-digit terms and which is therefore a product map.*

Basis encoding deserves a sentence, because it is the case that can be mistaken for a success. It attains $v^* = n$, outside the hypothesis of Theorem 4.1: its kernel is the identity, its induced distance is the discrete metric, and it satisfies Eq. (2) with zero violations while resolving nothing. That is why Section 3.1 insists on reporting resolution depth alongside violations.

Remark 4.4 (What this theorem does not cover). A ZZ or IQP feature map applies all-pairs couplings between qubits, so its state does not factorise across digits and Eq. (4) fails. Theorem 4.1 says nothing about it. We do not patch the gap by weakening the claim; we close it in Section 5 with an argument that ignores the structure of the encoding entirely.

Numerical verification. In all trials — five seeds at each of $(p, n) \in \{(2, 3), (3, 2), (2, 4), (5, 2)\}$ — randomly drawn product maps failed level constancy (`test_theorem_a_random_product_maps_are_never_str`). A product map built to satisfy the conclusion, with one non-trivial level and the rest trivial, yields exactly two distinct kernel values as predicted (`test_theorem_a_ultrametric_product_map_resolves_only_o`) and switching on a second non-trivial level destroys level constancy, which is the contradiction the proof derives (`test_theorem_a_a_nontrivial_deep_factor_destroys_ultrametricity`).

5. A Dimension Lower Bound for Arbitrary Encodings

Theorem 4.1 exploits the structure of the encoding. The bound in this section uses none: it constrains any map whatsoever into a space of given dimension, and so applies to entangling encodings, hardware-efficient ansätze, and designs not yet proposed.

Theorem 5.1 (Dimension lower bound). *Let $|X| = L < \infty$ and let $\Phi : X \rightarrow \mathbb{C}^D$ send its points to unit vectors with $|\langle \Phi(x) | \Phi(y) \rangle|^2 = K(x, y)$. Then*

$$D \geq \text{rank } G \geq \frac{L^2}{\sum_{x,y} K(x, y)}, \quad (7)$$

where G is the Gram matrix of the $\Phi(x)$.

Proof. The vectors $\Phi(x)$ lie in $\mathbb{C}^D$, so $\text{rank } G \leq D$. Let $r = \text{rank } G$ and let $\mu_1, \dots, \mu_r$ be the non-zero eigenvalues of G , which are real and non-negative because G is positive semi-definite. Since $\|\Phi(x)\| = 1$ we have $G_{xx} = 1$ and hence $\text{tr } G = \sum_k \mu_k = L$. By Cauchy–Schwarz applied to the vector of eigenvalues,

$$\|G\|_F^2 = \sum_{k=1}^r \mu_k^2 \geq \frac{1}{r} \left(\sum_{k=1}^r \mu_k \right)^2 = \frac{L^2}{r}. \quad (8)$$

Finally $\|G\|_F^2 = \sum_{x,y} |G_{xy}|^2 = \sum_{x,y} K(x, y)$ by the definition of the fidelity kernel. Rearranging Eq. (8) gives Eq. (7). $\square$

The bound is a statement about how *concentrated* the kernel is. A kernel close to the all-ones matrix has $\sum_{x,y} K \approx L^2$ and demands only one dimension, correctly: a near-constant kernel is realisable by a single state. A kernel close to the identity has $\sum_{x,y} K \approx L$ and demands L dimensions, also correctly. Strictly monotone ultrametric profiles decay fast enough to sit near the second regime, and the following corollary makes that quantitative.

Corollary 5.2 (No n -qubit encoding suffices for $p \geq 3$). *On the regular p -ary tree of depth n with the geometric profile $f(v) = p^{-(n-v)s}$, put*

$$S(p, s, n) = 1 + \frac{p-1}{p} \sum_{m=1}^n p^{m(1-s)}. \quad (9)$$

Then any exact realisation needs $D \geq p^n / S(p, s, n)$, that is, at least $n \log_2 p - \log_2 S(p, s, n)$ qubits. Fix $s > 1$. The sum then converges and $S(p, s, n) \leq S(p, s) < \infty$ uniformly in n , so the requirement grows like $n \log_2 p$ and exceeds n as soon as

$$n > n_0(p, s) = \frac{\log_2 S(p, s)}{\log_2 p - 1}. \quad (10)$$

For $p \geq 3$ the denominator is positive, so beyond that threshold no n -qubit encoding of any kind realises this kernel on $\mathbb{Z}/p^n$.

Proof. Fix a leaf x . The number of leaves y with $\lambda(x, y) = v$ is $p^{n-v} - p^{n-v-1}$ for $v < n$, and 1 for $v = n$. Hence $\sum_y f(\lambda(x, y)) = S(p, s, n)$ after substituting $m = n - v$, and by symmetry $\sum_{x,y} K = p^n S(p, s, n)$. Inserting this and $L = p^n$ into Eq. (7) gives the bound. For $s > 1$ the geometric series $\sum_m p^{m(1-s)}$ converges, so S is bounded uniformly in n . $\square$

Example 5.3. At $p = 3$, $s = 2$ we get $S \rightarrow 4/3$, so $n_0 = \log_2(4/3)/(\log_2 3 - 1) \approx 0.71$ and the requirement $q \geq 1.585n - 0.415$ exceeds n for every $n \geq 1$. At $n = 5$ the bound is 7.51 qubits against the 5 an angle or ZZ encoding provides; at $p = 5$ it is 11.3. Figure 3(b) plots the crossing.

The threshold is not decorative. At $p = 3$, $s = 1.01$ we get $S \approx 68$ and $n_0 \approx 7.2$, so the corollary says nothing for small trees — and indeed the bound at $n = 1$ is 0.85 qubits, while three qubit states with pairwise fidelity $3^{-1.01} \approx 0.33$ exist on the Bloch sphere. A *single* qubit realises that strictly monotone ultrametric kernel exactly. The hypothesis $n > n_0(p, s)$ is therefore necessary, not cosmetic.

Remark 5.4 (Scope). Corollary 5.2 is a statement about the geometric profile with parameter s , not about strictly monotone profiles in general. A profile close to constant has $\sum_{x,y} K \approx L^2$, so Eq. (7) degrades to $D \geq O(1)$ and narrow encodings are not excluded. We do not know whether they exist in that regime.

Remark 5.5 (The hypothesis $s > 1$ is load-bearing). At $s = 1$ every term of Eq. (9) equals 1 and $S(p, 1, n) = 1 + n(p - 1)/p$ grows with the depth, so the bound degrades to $p^n/\Theta(n)$ and Corollary 5.2 no longer follows from it. This is not an artefact of the proof: a slowly decaying profile really is easier to represent, because it is closer to constant. We record the boundary rather than quietly assuming it away (`test_theorem_b_requires_s_greater_than_one`).

Remark 5.6 (Theorems A and B are independent). At $p = 2$ the denominator of Eq. (10) vanishes: the bound reads $q \geq n - \log_2 S(2, s)$, which is below n for every s and every n ($\log_2 S(2, 2) \approx 0.58$). Theorem 5.1 therefore leaves n -qubit encodings on binary trees open, and it is Theorem 4.1's structural argument that closes the product case there. Conversely Theorem 4.1 says nothing about entangling maps at any radix. Neither theorem subsumes the other, and we present them as covering different regimes rather than as a chain (`test_theorem_b_does_not_subsume_theorem_a_at_radix_two`).

Numerical verification. The closed form Eq. (9) matches the empirical row sum to machine precision (`test_theorem_b_closed_form_row_sum`); the bound exceeds 2^n for $(p, n) \in \{(3, 2), (3, 3), (3, 4), (5, 2)\}$ (`test_theorem_b_rules_out_every_n_qubit_encoding`); and the ZZ feature map, which lives in 2^n dimensions, falls below the bound and indeed fails to reproduce the profile (`test_theorem_b_zz_feature_map_is_below_the_bound`).

Two cautions on applying Eq. (7) in practice. Evaluated on a subsample of L' leaves it remains valid — restricting the point set can only make the kernel easier to realise — but it is capped at L' , so a subsampled estimate understates the bound and saturates at $\log_2 L'$ qubits. The figures in Section 8 therefore use the closed form. And the bound constrains dimension, not circuit depth: it says nothing about how hard the required state is to prepare, which is the subject of Section 6.

6. Exact Realisation by Path States

The two obstructions leave a gap: strictly monotone ultrametric kernels need many dimensions and cannot be factorised across digits. The construction in this section fills the gap exactly, and turns out to characterise the whole family.

Theorem 6.1 (Path states). *Let T be a finite rooted tree on node set V with all leaves at depth n . For every strictly increasing $f : \{0, \dots, n\} \rightarrow (0, 1]$ with $f(n) = 1$, set*

$$a_i^2 = \sqrt{f(i)} - \sqrt{f(i-1)} \quad (0 \leq i \leq n), \quad \sqrt{f(-1)} := 0, \quad (11)$$

and define the path state

$$\Phi_f(x) = \sum_{i=0}^n a_i |\text{anc}_i(x)\rangle \in \mathbb{C}^V. \quad (12)$$

Then $\Phi_f(x)$ is a unit vector and, for all leaves x, y ,

$$|\langle \Phi_f(x) | \Phi_f(y) \rangle|^2 = f(\lambda(x, y)) \quad \text{exactly.} \quad (13)$$

Conversely, every kernel that is ultrametric with a strictly monotone profile arises this way, and Φ_f is the unique realisation up to a global isometry and a phase on each point among realisations whose pairwise overlaps are real and non-negative. Dropping that restriction, uniqueness fails (Remark 6.2).

Proof. Strict monotonicity makes each a_i^2 in Eq. (11) positive, so the a_i are well defined and real. The sum telescopes, $\sum_{i=0}^n a_i^2 = \sqrt{f(n)} = 1$, so $\Phi_f(x)$ is a unit vector.

For the overlap, the key structural fact is that two leaves share precisely their ancestors down to the depth at which they meet, and no deeper: $\text{anc}_i(x) = \text{anc}_i(y)$ if and only if $i \leq \lambda(x, y)$. The basis vectors $|u\rangle$ for $u \in V$ being orthonormal,

$$\langle \Phi_f(x) | \Phi_f(y) \rangle = \sum_{i=0}^n a_i^2 \mathbf{1}[\text{anc}_i(x) = \text{anc}_i(y)] = \sum_{i \leq \lambda(x, y)} a_i^2 = \sqrt{f(\lambda(x, y))}, \quad (14)$$

again by telescoping. Squaring the modulus gives Eq. (13).

Two remarks on Eq. (14). The cross terms vanish because $\text{anc}_i(x)$ and $\text{anc}_j(y)$ sit at different depths when $i \neq j$, so they are distinct nodes and hence orthogonal basis vectors. And $a_i^2 = \sqrt{f(i)} - \sqrt{f(i-1)}$ is forced by Eq. (14) read at successive levels, so f determines the amplitudes.

For the converse, let Ψ be a realisation whose Gram matrix G^Ψ has real non-negative entries. Then $G_{xy}^\Psi = \sqrt{f(\lambda(x, y))}$ exactly, which by Eq. (14) is the Gram matrix of Φ_f . Two families of vectors with equal Gram matrices are related by an isometry of their spans, which is the standard argument; composing with per-point phases covers the residual freedom that leaves $|\cdot|$ unchanged. $\square$

Remark 6.2 (How far uniqueness actually goes). Two weakenings are needed, and the second is not cosmetic.

First, the converse cannot say “unique up to relabelling of V ”. Rotating every $\Phi_f(x)$ by a fixed unitary, or multiplying each by its own phase, leaves the fidelity kernel unchanged while destroying the reading of the basis as tree nodes (`test_theorem_c_uniqueness_up_to_isometry_and_phase`). The node-labelled form Eq. (12) is the canonical representative of its isometry class, not the only member.

Second, and less obviously, the positivity hypothesis cannot be dropped. A fidelity kernel fixes only the *moduli* $|G_{xy}|$, and these do not determine G up to phases: the Bargmann invariants $G_{xy}G_{yz}G_{zx}$ are invariant under $\Psi(x) \mapsto e^{i\vartheta_x}U\Psi(x)$ yet are not functions of the moduli. Concretely, take $n = 1$, $p = 3$, $f(0) = t = 1/4$ and

$$G^\Psi = \begin{pmatrix} 1 & \sqrt{t} & i\sqrt{t} \\ \sqrt{t} & 1 & \sqrt{t} \\ -i\sqrt{t} & \sqrt{t} & 1 \end{pmatrix},$$

Table 1

Resources for the path-state encoding, and the Theorem 5.1 bound it must respect. Dataset rows use the depth-8 truncation of Section 8; the WordNet row is the full hierarchy before truncation.

	nodes $ V $	qubits	non-zero amplitudes	preparation gates
general tree	$ V $	$\lceil \log_2 V \rceil$	$n + 1$	$O(n \log V)$
$\mathbb{Z}/p^n$	$\frac{p^{n+1}-1}{p-1}$	$\approx n \log_2 p$	$n + 1$	$O(n^2 \log p)$
WordNet (full)	74 374	17	$19 + 1$	—
WordNet (truncated)	1 028	11	$8 + 1$	—

which is positive semi-definite with spectrum $\{0.134, 1, 1.866\}$ and realises the same kernel as Φ_f . Its Bargmann invariant is $-\frac{1}{8}i$ against $+\frac{1}{8}$ for the path state, so no isometry and no choice of phases carries one to the other. The correspondence $f \leftrightarrow \Phi_f$ is therefore a bijection onto strictly monotone ultrametric kernels only after restricting to realisations with non-negative overlaps (`test_theorem_c_converse_needs_non_negative_overlaps`).

Remark 6.3 (Positive semi-definiteness for free). Let A be the leaf-by-node incidence matrix with $A_{xu} = a_{\text{depth}(u)}$ if u is an ancestor of x and 0 otherwise. Then Eq. (14) says $\sqrt{K} = AA^\top$ entrywise, so $K = (AA^\top)^{\circ 2}$ is a Schur square of a positive semi-definite matrix and hence positive semi-definite. This re-derives, in one line, the classical fact that monotone ultrametric kernels are positive semi-definite.

6.1. Resources

The path state lives in $\mathbb{C}^V$, so it needs $\lceil \log_2 |V| \rceil$ qubits. Critically, it has only $n + 1$ non-zero amplitudes out of $|V|$, so it is a sparse state. In a level-indexed layout, where the register splits into a depth counter and a within-level index, preparation is a cascade of n controlled rotations — one per level, each splitting the remaining amplitude between “stop here” and “descend” — followed, at each level, by writing the within-level index of $\text{anc}_i(x)$ conditioned on the depth register. Counting both parts gives $O(n \log |V|)$ elementary gates. We quote that figure throughout; the $O(n)$ rotations alone are not the whole cost.

Table 1 collects the counts. The headline is that the full WordNet noun hierarchy — 74 374 nodes — fits in 17 qubits, with $19 + 1$ non-zero amplitudes.

Optimality. Comparing with Corollary 5.2: on $\mathbb{Z}/p^n$ the construction uses $\log_2 |V| = n \log_2 p + \log_2 \frac{p}{p-1} + o(1)$ qubits, while the bound demands at least $n \log_2 p - \log_2 S(p, s)$. The two differ by an additive constant independent of n , so the path-state encoding is optimal up to $O(1)$ qubits. We resist the stronger phrasing: it is not exactly optimal, and the gap is a genuine, if small, one.

6.2. The p -adic specialisation

On the regular p -ary tree, $\lambda(x, y) = \min(v_p(x - y), n)$ and the profile can be tied directly to the p -adic absolute value. With $|x - y|_p = p^{-\lambda}$ for $x \neq y$, the geometric profile is

$$f(\lambda) = p^{-(n-\lambda)s} = \left(\frac{p^{-n}}{|x - y|_p} \right)^s, \quad (15)$$

so the kernel is a decreasing power of the p -adic distance, with $f(n) = 1$ filling in the value at $x = y$ where $|x - y|_p = 0$. The path state is then a level-weighted superposition over the root-to-leaf path in the p -adic tree, with amplitude concentrated near the leaf for large s and spread towards the root for small s . This is the same object that appears as the state space of continuous-time quantum walks on p -adic trees, where the vertices are sub-balls of a p -adic ball [4, 5], and it connects the encoding to the p -adic quantum formalism of Aniello et al. [6].

6.3. Arbitrary rooted trees

Nothing in the proof of Theorem 6.1 uses regularity. The only structural input is Eq. (14), which needs just that ancestors nest — true in any rooted tree. Branching factors may vary between nodes and between levels, and shallow leaves are handled by the unary padding of Section 3, which changes no lowest common ancestor. The construction therefore applies verbatim to WordNet, whose branching factor ranges from 1 to 402, and $\mathbb{Z}/p^n$ is the homogeneous specialisation rather than the setting (`test_theorem_c_generalises_to_a_non_homogeneous_tree`).

Numerical verification. Across $(p, n) \in \{(2, 3), (3, 2), (2, 4), (5, 2)\}$ and both geometric and linear profiles, the residual $\max_{x,y} |K - f(\lambda)|$ is at most 2.22×10^{-16} , and the induced distance has zero strong-triangle violations (`test_theorem_c_path_states_realise_the_profile_exactly`, `test_theorem_c_induced_distance_is_exactly_ultrametric`).

7. Beyond Exactness: Re-uploading and Noise

7.1. Why an extension must re-upload the data

Theorem 6.1 gives a fixed encoding with no trainable parameters. The obvious way to add expressivity is to follow it with a parametrised circuit, but that does nothing at all: for any unitary $W(\theta)$,

$$|\langle \Phi_f(x) | W(\theta)^\dagger W(\theta) | \Phi_f(y) \rangle|^2 = |\langle \Phi_f(x) | \Phi_f(y) \rangle|^2, \quad (16)$$

so the kernel is invariant. Any non-trivial extension must therefore re-upload the data between layers [7, 8]. We use the family

$$\Phi^{(L)}(x) = W_L D_\gamma(x) \cdots W_1 D_\gamma(x) \Phi_f(x), \quad (17)$$

where $D_\gamma(x)$ applies the phase $e^{i\gamma}$ to every node on the root-to-leaf path of x — the natural data-dependent diagonal on the path-state Hilbert space — and each W_ℓ interpolates between the identity and a Haar-random unitary by a parameter we call the entangling strength. At $\gamma = 0$ with trivial mixers, Eq. (17) reduces exactly to $\Phi_f(x)$, so the exact construction sits at the origin of the family and anchors any trade-off curve at zero distortion. Note that the innermost $D_\gamma(x)$ phases exactly the support of $\Phi_f(x)$ and so acts as a global phase; the family is non-trivial only because the mixers rotate the state between successive re-uploadings.

We measure two quantities: the *ultrametric distortion*

$$\mathcal{D} = \max_{x,y,z} \max(0, d(x, z) - \max(d(x, y), d(y, z))), \quad d = 1 - K, \quad (18)$$

which vanishes exactly when the induced distance is an ultrametric, and the kernel-target alignment as a standard proxy for usable expressivity. Section 8 maps the resulting Pareto front.

7.2. Depolarising noise preserves the strong triangle inequality

Exactness is a statement about an ideal kernel, so it is fair to ask what survives on hardware. For the dominant global error channel, the geometry survives intact and only the contrast degrades.

Proposition 7.1 (Depolarising noise preserves the strong triangle inequality). *Let K be ultrametric with non-decreasing profile f , realised on a Hilbert space of dimension $D_{\mathcal{H}} = 2^{\lceil \log_2 |V| \rceil}$, and let $K_r = (1 - r)K + r/D_{\mathcal{H}}$ be the kernel estimated under global depolarising noise of rate $r \in [0, 1]$. Then K_r is level-constant with non-decreasing profile $(1 - r)f + r/D_{\mathcal{H}}$, and $d_r = 1 - K_r$ satisfies Eq. (2) for every r .*

Table 2

The four hierarchies after preprocessing. Source columns describe the full reduced tree; the remaining columns describe the sample the kernels are evaluated on. Qubits is $\lceil \log_2 |V| \rceil$ for the path-state encoding.

dataset	source nodes	source height	sampled leaves	nodes	radix	qubits
synthetic 2-ary	511	8	256	511	2	9
WordNet	74 374	19	256	1 028	11	11
Gene Ontology	10 041	12	256	1 235	16	11
NCBI Mammalia	14 722	14	256	633	38	10

Proof. The compute-uncompute estimator reads off the probability of the all-zeros outcome; replacing the state by the maximally mixed state with probability r sends that probability to $1/D_{\mathcal{H}}$, which gives the stated K_r . Hence

$$d_r = 1 - (1 - r)K - \frac{r}{D_{\mathcal{H}}} = (1 - r)d + r\left(1 - \frac{1}{D_{\mathcal{H}}}\right),$$

an affine function $\varphi(t) = (1 - r)t + c$ of d with non-negative slope. A non-decreasing affine map preserves order and commutes with $\max$, so $d(x, z) \leq \max(d(x, y), d(y, z))$ implies $\varphi(d(x, z)) \leq \max(\varphi(d(x, y)), \varphi(d(y, z)))$, which is Eq. (2) for d_r . Applying φ to $K = f(\lambda)$ gives the stated profile, which is still non-decreasing. $\square$

Remark 7.2 (Why this is not quite “ K_r is ultrametric”). For $r > 0$ the profile of K_r has value $(1 - r) + r/D_{\mathcal{H}} < 1$ at $v = n$, so it violates the normalisation $f(n) = 1$ of Definition 3.1, and correspondingly $d_r(x, x) > 0$: strictly, d_r is a semi-metric rather than a metric. What the proposition establishes is that the two properties we actually measure — level constancy and the strong triangle inequality — both survive. Rescaling K_r by its diagonal restores $f(n) = 1$ and changes nothing else, since the rescaling is by a constant.

What depolarising noise destroys is not geometry but contrast. The profile is compressed towards the constant $1/D$, so consecutive levels become harder to tell apart and the resolution depth of Definition 3.3 falls, even though level constancy stays exactly zero. Section 8 measures where that collapse happens.

Sampling noise behaves in the opposite way, and the reason is specific to ultrametric spaces. Because every triangle in an ultrametric is isocles with its two longest sides equal, a constant fraction of triples — 0.667 of them on the tree used in our noise experiment — satisfies Eq. (2) *with equality*. An arbitrarily small independent perturbation flips about half of those into violations. Counting violations is therefore an unstable statistic at low noise: it saturates rather than decaying as the shot budget grows. The violation *magnitude* is the meaningful quantity, and it decays as $O(1/\sqrt{\text{shots}})$ as one would expect. We return to this in Section 8.6.

8. Experiments

All experiments are analytic or statevector simulations on CPU, using NumPy and scikit-learn, and each completes in seconds to minutes. Seeds, resolved configurations and library versions are recorded alongside every run. Every number quoted below is generated from the experiment outputs rather than transcribed by hand.

8.1. Setup

Datasets. We use WordNet nouns via NLTK, the molecular-function namespace of the Gene Ontology, the Mammalia clade of the NCBI taxonomy, and synthetic regular p -ary trees for controlled ablations. Table 2 gives the shapes.

Table 3

E1. Violation rate counts triples breaking Eq. (2); level constancy is zero exactly when the kernel is ultrametric with some profile; resolution is the number of distinguishable kernel values against the number of lowest-common-ancestor levels the sample populates. Only the path state scores well on both.

dataset	encoding	violation rate	level constancy	resolution
synthetic 2-ary	path state (ours)	0	0	9 / 9
	basis	0	0	2 / 9
	angle	0.234	0.496	9 / 9
	ZZ / IQP	0.275	0.983	9 / 9
	random product	0.329	0.927	9 / 9
WordNet	path state (ours)	0	4.44×10^{-16}	9 / 9
	basis	0	0	2 / 9
	angle	0.322	0.978	9 / 9
	ZZ / IQP	0.328	0.48	9 / 9
	random product	0.329	0.991	9 / 9
Gene Ontology	path state (ours)	0	4.44×10^{-16}	9 / 9
	basis	0	0	2 / 9
	angle	0.322	0.986	9 / 9
	ZZ / IQP	0.327	0.748	9 / 9
	random product	0.329	0.992	9 / 9
NCBI Mammalia	path state (ours)	0	4.44×10^{-16}	9 / 9
	basis	0	0	2 / 9
	angle	0.323	0.998	9 / 9
	ZZ / IQP	0.328	0.605	9 / 9
	random product	0.329	0.999	9 / 9

Preprocessing, stated rather than assumed. Each of these sources is a directed acyclic graph in the wild, not a tree: WordNet synsets can have several hypernyms. We reduce each to a tree by keeping, for every node, the *longest* path to the root, breaking ties by lexicographically smallest identifier. Multiple inheritance is therefore discarded, which we record as a limitation in Section 10 rather than as a detail.

We then truncate every hierarchy at depth 8. This is needed because the 2^{height} -dimensional product baselines are otherwise not representable: WordNet reaches depth 19, so an angle or ZZ encoding would require 2^{19} amplitudes per point. Truncation also has the further benefit that every encoding is compared on exactly the same tree. Finally we sample 256 leaves, restrict to their ancestor closure, and pad shallow leaves to uniform depth. None of these steps changes the lowest-common-ancestor depths among the sampled leaves below the truncation, which is all the kernel depends on.

Baselines. Angle encoding, computational-basis encoding, a ZZ/IQP feature map with two repetitions, a random product map, a block-product map where representable, and two classical references: an RBF kernel on integer leaf coordinates and one on a one-hot encoding. The geometric profile uses base 2 throughout for the real datasets. Tying the profile base to the branching factor instead, 38 for NCBI, pushes $f(0)$ to around 10^{-13} , below any resolution tolerance that a realistic shot budget could support, so the base is treated as the free design parameter it is.

8.2. E1: ultrametricity and resolution

Table 3 reports both diagnostics of Definition 3.3 across all four hierarchies, and Figure 2 plots them for WordNet.

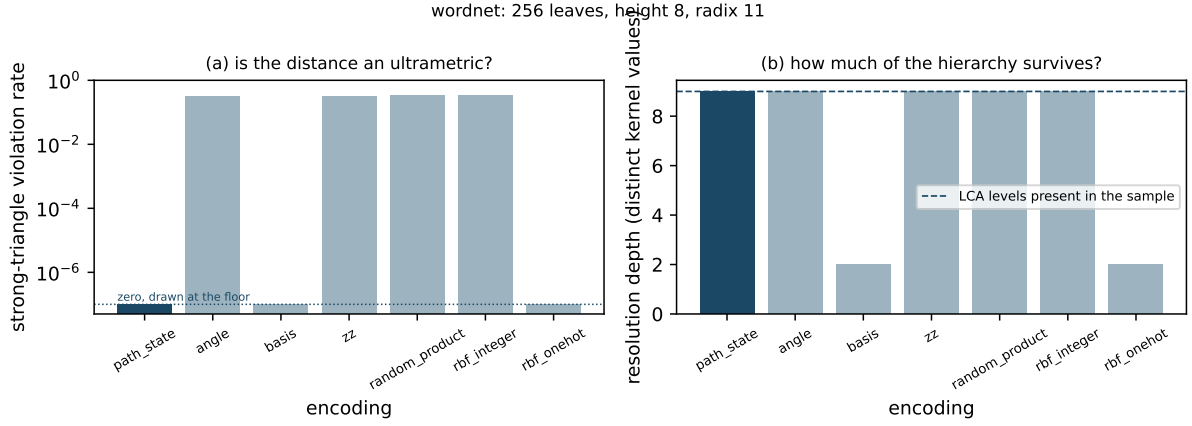


Figure 2: E1 on WordNet. Panel (a) asks whether the induced distance is an ultrametric at all; panel (b) asks how much of the hierarchy survives. Basis encoding passes (a) perfectly and fails (b) completely, which is why both are reported.

Three readings. First, the path-state encoding attains level constancy 4.44×10^{-16} and zero violations on every dataset, with full resolution: the construction is exact on real hierarchies, not only on the regular trees of Theorem 6.1. Second, every product and entangling baseline has a level constancy near 1: angle reaches 0.978 and the random product map 0.991. Their kernels are not merely imperfect ultrametrics but are not ultrametric at all, in agreement with Theorems 4.1 and 5.1.

Third, and worth stating plainly because it complicates the headline: **basis encoding also achieves zero violations**, with level constancy exactly zero. Its resolution is 2 out of 9. This is the degenerate corner of Theorem 4.1 discussed in Section 4, and it is the reason we do not claim that the path state is the unique encoding with an ultrametric kernel. The correct claim is narrower and, we think, more useful: it is the only one that is both ultrametric and fully resolving.

8.3. E2: hierarchical classification

Table 4 reports five-fold cross-validated accuracy for predicting a leaf’s depth-2 ancestor, together with the Spearman correlation between the kernel’s induced distance and true tree distance. Classes with fewer than five members are dropped and the count is reported in the run metadata; no hyperparameter was tuned.

The path state attains $\rho = 1$ on every dataset. This is not a fitted result and should not be read as one: by Eq. (13) the induced distance is a strictly monotone function of tree distance, so perfect rank correlation is a consequence of the construction. No baseline exceeds 0.368.

On accuracy the picture is different, and we report it as it came out. On WordNet the path state reaches 0.792 ± 0.0157 while the random product baseline reaches 0.839; the ordering is similar on Gene Ontology. A kernel that faithfully encodes tree distance is therefore *not* the best kernel for this classification task, which is unsurprising once stated: the label is the depth-2 ancestor, and a kernel that also spends representational capacity on depths 3 through 8 is spending it on distinctions the label ignores. Geometric fidelity and task accuracy are different objectives, and this paper claims the former.

The comparison against v-PuNNs [9], which reports 99.96% leaf accuracy on WordNet, must be read the same way and is not close. That is a trained deep model with a bespoke optimiser over p -adic weight space; this is a fixed encoding with a closed form and no parameters. We record the gap rather than reframing it: on accuracy, the trained model wins by a wide margin. The criterion for this experiment was fixed before any number was produced, precisely so that a weak result could not tempt a retrofit of the paper’s claims.

Table 4

E2. Accuracy is five-fold cross-validated with a precomputed-kernel SVM at $C = 1$. A dash marks an undefined rank correlation: a delta kernel assigns every distinct pair the same distance, so no correlation exists.

dataset	encoding	leaf accuracy	Spearman ρ
synthetic 2-ary	path state (ours)	1 ± 0	1
	angle	1 ± 0	0.419
	ZZ / IQP	0.722 ± 0.133	0.173
	random product	0.379 ± 0.0811	0.0939
	basis	0.238 ± 0.00588	—
	RBF (integer)	0.188 ± 0.03	-0.0122
WordNet	path state (ours)	0.792 ± 0.0157	1
	angle	0.675 ± 0.0293	0.175
	ZZ / IQP	0.478 ± 0.0319	0.098
	random product	0.839 ± 0.056	0.368
	basis	0.212 ± 0.00784	—
	RBF (integer)	0.145 ± 0.0157	0.00375
Gene Ontology	path state (ours)	0.601 ± 0.0357	1
	angle	0.561 ± 0.0316	0.135
	ZZ / IQP	0.483 ± 0.0263	0.157
	random product	0.75 ± 0.0249	0.283
	basis	0.281 ± 0.00742	—
	RBF (integer)	0.272 ± 0.018	0.027
NCBI Mammalia	path state (ours)	1 ± 0	1
	angle	0.85 ± 0.0365	-0.00603
	ZZ / IQP	0.897 ± 0.0609	0.101
	random product	1 ± 0	0.349
	basis	0.677 ± 0.00888	—
	RBF (integer)	0.677 ± 0.00888	0.0645

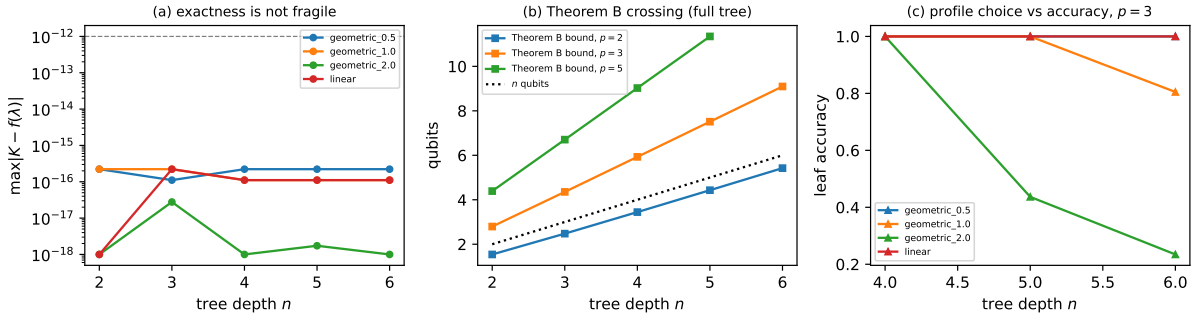


Figure 3: E3. (a) The profile residual stays at machine precision for every strictly monotone profile at every depth. (b) The Theorem 5.1 bound against the n qubits a product or IQP encoding provides: above the diagonal for $p \geq 3$, below it for $p = 2$. (c) Profile choice against accuracy.

8.4. E3: ablations

Figure 3 sweeps the profile shape, tree depth and branching factor over 56 configurations.

Panel (a) is the robustness check for Theorem 6.1: the worst residual over all 56 configurations is 2.22×10^{-16} , and the total number of strong-triangle violations is 0. Exactness does not degrade with depth, radix or profile shape.

Panel (b) is the picture of Corollary 5.2. For $p = 3$ and $p = 5$ the required qubit count runs above the diagonal at every depth, so no n -qubit encoding can realise the kernel. For $p = 2$ it runs below, which is Remark 5.6: at radix two the dimension bound alone is not enough and

wordnet: 96 leaves, height 8

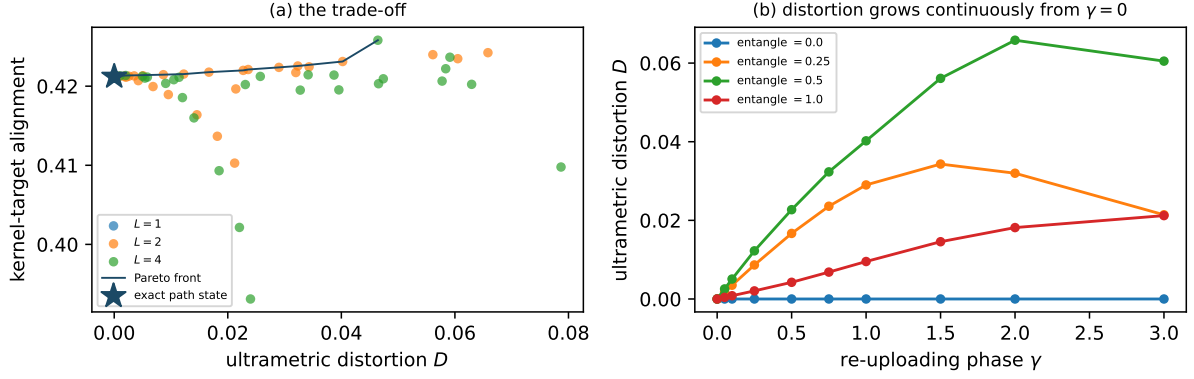


Figure 4: E4 on WordNet, over 120 configurations of the family Eq. (17). Distortion grows continuously from zero, so the exact construction is the endpoint of a tunable family rather than an isolated point — but the alignment available in exchange is small.

synthetic: 32 leaves, height 5, dim 63

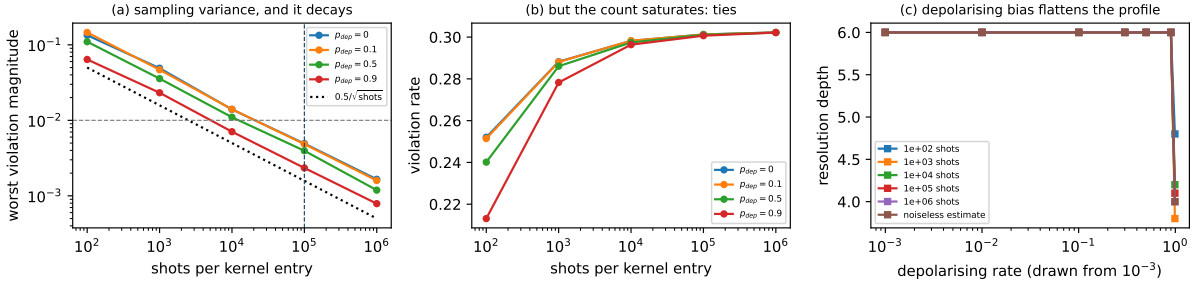


Figure 5: E5. (a) The violation magnitude decays as $O(1/\sqrt{\text{shots}})$, tracking the dotted reference line. (b) The violation *count* saturates instead, because a constant fraction of triples in an ultrametric is exactly tight. (c) Depolarising noise leaves ultrametricity intact but flattens the profile until resolution collapses.

Theorem 4.1 carries the argument. The curves use the closed form Eq. (9) on the full tree, since an estimate from a subsample saturates at $\log_2$ of the sample size.

8.5. E4: the re-uploading trade-off

Distortion grows continuously from zero as the re-uploading phase increases, which answers the structural question: the exact construction is not an isolated special case but the zero-distortion endpoint of a continuum. The quantitative answer is less encouraging. The exact point achieves alignment 0.421, and the best configuration anywhere in the family achieves 0.426 at distortion 0.0464, a gain of 0.00459. On these datasets the trade-off exists but buys very little, and we would not recommend paying for it.

8.6. E5: noise and finite shots

Proposition 7.1 predicts that depolarising noise cannot break ultrametricity, and the measurement agrees exactly: at every rate up to 0.99, the noiseless-estimate row has zero violations and zero excess. What degrades is resolution, and only at extreme rates.

Finite sampling is the real constraint, and measuring it correctly required abandoning the obvious statistic. The violation count *increases* with the shot budget, rising towards a plateau, which looks like a bug and is not: 0.667 of triples in this ultrametric meet Eq. (2) with equality, and any perturbation flips about half of them, so the count converges to that tie fraction rather

than to zero. The violation magnitude behaves as expected, falling from 0.134 at 100 shots to 0.00165 at 1 000 000 shots, a factor of $\sqrt{10}$ per decade. Defining the budget on the magnitude gives 100 000 **shots per kernel entry** for a worst-case excess below 0.01, over 10 repetitions.

9. Related Work

***p*-adic quantum mechanics.** Quantum systems over *p*-adic fields have a developed formalism. Aniello et al. [6] construct a model of quantum states over a quadratic extension of $\mathbb{Q}_p$, working the *p*-adic qubit and quNit out explicitly. Zúñiga-Galindo [4] and Zúñiga-Galindo [5] study continuous-time quantum walks whose vertex set consists of the sub-balls of a *p*-adic ball, that is, on the nodes of the *p*-adic tree, and Zúñiga-Galindo and Chacón-Cortés [10] relate discretised *p*-adic Schrödinger dynamics to continuous-time quantum Markov chains. Our path states live in $\mathbb{C}^V$ with V the node set of that same tree, so the encoding of Theorem 6.1 is a state in the Hilbert space these works take as their setting. We use it statically, as a feature map, rather than evolving it; connecting the profile f to the spectrum of a *p*-adic walk generator is a natural next question that we do not address.

***p*-adic machine learning.** The closest prior work is v-PuNNs [9], which builds neurons as characteristic functions of *p*-adic balls, proves a finite hierarchical approximation theorem, and reports strong results on WordNet, Gene Ontology and the NCBI taxonomy. Our benchmarks are taken from that work so the comparison is direct. The relationship is complementary rather than competitive: v-PuNNs is a trained architecture over *p*-adic weight space, while this paper characterises which *quantum encodings* can carry ultrametric structure at all. v-PuNNs is accordingly far more accurate on the shared benchmarks, as we report in Section 8.3. Separately, Bradley [3] asks how ultrametric real finite metric data actually is, recasting Vietoris–Rips complexes via *p*-adic Mumford curves; that question is prior to ours, which assumes the target geometry is ultrametric and asks what can represent it.

Quantum kernels. The fidelity kernel construction is due to Havlíček et al. [2], and Schuld [1] showed that supervised quantum models are kernel methods, which is what makes the encoding the locus of inductive bias. A recent line of work establishes limits on what these kernels can do: exponential concentration as system size grows [11], double descent [12] and benign overfitting [13] in their generalisation behaviour, and trainable embeddings as a response [8]. Those results concern how a quantum kernel behaves once chosen. Theorems 4.1 and 5.1 concern which kernels are reachable in the first place, and are in that sense upstream: our obstructions hold before any question of trainability or concentration arises. Theorem 5.1 in particular does not care whether the encoding concentrates, only how wide it is.

Hierarchy in classical representation learning. Hyperbolic embeddings are the established answer to representing trees [14], and the approach has grown into a substantial literature [15]. Hyperbolic space achieves low but non-zero distortion of tree metrics, because a tree embeds into it only approximately. The path-state construction achieves zero distortion by design, at the cost of a dimension linear rather than logarithmic in the number of nodes. By Theorem 5.1, that cost is not an artefact of our construction but a lower bound for any exact realisation.

Data re-uploading. Interleaving data encodings with trainable layers is due to Pérez-Salinas et al. [7] and remains actively developed [16]. We use it in Section 7 not to increase expressivity for its own sake but as the minimal way to extend a fixed encoding at all, given the invariance Eq. (16).

10. Limitations and Outlook

No quantum speedup in kernel evaluation. Restating the point of Section 1 where a reader will look for it: $K(x, y) = f(\lambda(x, y))$ is computable classically in time linear in the tree depth, so nothing here accelerates kernel evaluation. The obstructions of Theorems 4.1 and 5.1 constrain quantum encoding design, and the construction of Theorem 6.1 is useful as an input stage for deeper quantum models, where the closed form does not survive subsequent layers. Neither is a speedup claim.

Multiple inheritance is discarded. All three real hierarchies are directed acyclic graphs, and the longest-path reduction of Section 8.1 throws away every hypernym but one. A synset with two genuinely distinct parents is misrepresented. Extending the construction to DAGs is not a matter of a better reduction: the ancestor sets of two leaves in a DAG need not be nested, and Eq. (14) — the entire content of Theorem 6.1 — depends on that nesting.

Truncation. Comparing against 2^{height} -dimensional baselines forced a depth-8 truncation. The path-state encoding itself has no such limit — Table 1 gives 17 qubits for the untruncated WordNet hierarchy — so the truncation is a constraint of the comparison, not of the method. We have not measured the baselines at full depth, because they cannot be represented there.

The dimension bound is loose for flat profiles. Theorem 5.1 is strong when the profile decays quickly and weak when it does not, and Remark 5.5 shows the hypothesis $s > 1$ is necessary rather than convenient. For slowly decaying profiles the bound does not rule out narrow encodings, and we do not know whether narrow encodings actually exist in that regime.

Simulation only. Everything here is statevector simulation. Proposition 7.1 covers the global depolarising channel analytically, and Section 8.6 gives a shot budget of 100 000 per kernel entry, but coherent errors, readout error and device-specific noise are untested. The shot budget is also per entry, so a full $m \times m$ kernel matrix costs $O(m^2)$ times that, which is the practical barrier to using this at scale on hardware.

E2 is parity evidence, not a win. As reported in Section 8.3, the path state loses on classification accuracy to both a random product map and to v-PuNNs. The paper’s claims rest on Theorems 4.1, 5.1 and 6.1 and on E1, not on E2.

Outlook. Three directions seem worth pursuing. The most immediate is using the path state as the input layer of a deeper variational model, where Section 8.5 suggests the interesting regime is small distortion rather than large. The second is the p -adic connection: the encoding lives in the state space of quantum walks on the p -adic tree [4], and relating the profile f to the spectrum of a walk generator would give a dynamical reading of what is currently a static construction. The third is gate synthesis by p -adic descent, where the ultrametric structure would organise the search rather than the data; that is a longer project and we mention it only to place the present work.

11. Conclusion

The geometry a quantum kernel imposes on its data is decided by the encoding, before training begins. For hierarchical data that geometry should be ultrametric, and we have shown that the encodings in common use cannot make it so. Product feature maps fail for a structural reason — multiplicativity across digits is incompatible with resolving a tree beyond a bounded depth —

and encodings of insufficient width fail for a reason that does not depend on their structure at all, which is what rules out the entangling maps the first argument cannot reach.

The positive result is that path states close the gap exactly. They realise precisely the strictly monotone ultrametric kernels, on arbitrary rooted trees, using a number of qubits within an additive constant of the proven lower bound, with $n + 1$ non-zero amplitudes and $O(n)$ controlled rotations. Their ultrametricity is immune to global depolarising noise, and survives finite sampling at a measured budget of 100 000 shots per kernel entry.

What this does not deliver is a faster kernel. The construction's value is as a constraint on encoding design and as an input stage for models where no closed form survives, and as an argument that the encoding deserves to be chosen for the geometry of the data rather than for the convenience of the circuit.

Declaration on Generative AI

Complete per the CEUR-WS generative-AI disclosure policy in force at submission.

References

- [1] M. Schuld, Supervised quantum machine learning models are kernel methods, 2021. [arXiv:2101.11020](#).
- [2] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, *Nature* 567 (2019) 209–212. doi:10.1038/s41586-019-0980-2. [arXiv:1804.11326](#).
- [3] P. E. Bradley, On the local ultrametricity of finite metric data, *Journal of Classification* 42 (2025) 624–638. doi:10.1007/s00357-025-09508-3. [arXiv:2408.07174](#).
- [4] W. A. Zúñiga-Galindo, p-adic quantum mechanics, infinite potential wells, and continuous-time quantum walks, 2024. [arXiv:2410.13048](#).
- [5] W. A. Zúñiga-Galindo, 2-adic quantum mechanics, continuous-time quantum walks, and the space discreteness, *Fortschritte der Physik* (2025). doi:10.1002/prop.70019. [arXiv:2502.16416](#).
- [6] P. Aniello, S. Mancini, V. Parisi, A p-adic model of quantum states and the p-adic qubit, *Entropy* 25 (2023) 86. doi:10.3390/e25010086.
- [7] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, J. I. Latorre, Data re-uploading for a universal quantum classifier, *Quantum* 4 (2020) 226. doi:10.22331/q-2020-02-06-226. [arXiv:1907.02085](#).
- [8] P. Rodriguez-Grasa, Y. Ban, M. Sanz, Neural quantum kernels: training quantum kernels with quantum neural networks, 2024. [arXiv:2401.04642](#).
- [9] G. L. R. N'guessan, v-PuNNs: van der Put neural networks for transparent ultrametric representation learning, 2025. [arXiv:2508.01010](#), preprint; revised January 2026.
- [10] W. A. Zúñiga-Galindo, L. F. Chacón-Cortés, Continuous-time quantum markov chains and discretizations of p-adic Schrödinger equations: Comparisons and simulations, 2025. [arXiv:2508.06712](#).
- [11] S. Thanasilp, S. Wang, M. Cerezo, Z. Holmes, Exponential concentration in quantum kernel methods, *Nature Communications* 15 (2024) 5200. doi:10.1038/s41467-024-49287-w. [arXiv:2208.11060](#).
- [12] M. Kempkes, A. Ijaz, E. Gil-Fuster, C. Bravo-Prieto, J. Spiegelberg, E. van Nieuwenburg, V. Dunjko, Double descent in quantum kernel methods, 2025. [arXiv:2501.10077](#).
- [13] J. Tomasi, S. Anthoine, H. Kadri, Benign overfitting with quantum kernels, 2025. [arXiv:2503.17020](#).
- [14] M. Nickel, D. Kiela, Poincaré embeddings for learning hierarchical representations, 2017. [arXiv:1705.08039](#), advances in Neural Information Processing Systems 30.

- [15] N. He, H. Madhu, N. Bui, M. Yang, R. Ying, Hyperbolic deep learning for foundation models: A survey, 2025. [arXiv:2507.17787](#).
- [16] L. Cassé, B. Pfahringer, A. Bifet, F. Magniette, Quantum re-uploading for calorimetry: Optimized architectures with extended expressivity, 2024. [arXiv:2412.12397](#).